

EVIDENCE - BASED COACHING MANUAL

THE LEAN MACHINE PROTOCOL

A 10-Week Body Recomposition System — Built for You

81→75

KG TARGET

~22→15

% BODY FAT

10

WEEKS

4+1

DAY SPLIT

Profile: 28 y • 174 cm • Advanced • Vegetarian • AM Training • 10k steps/day

1 The Training Split Decision

28 yrs
Age

174 cm
Height

81 kg
Weight

~22%
Body fat

3+ yrs
Advanced

Vegetarian
+ dairy/whey

7-8 h
Sleep

10,000
Steps/day

YOUR SPLIT:

4-Day UPPER / LOWER (+ 1 optional weak-point day)

Scientific Comparison

Criterion	PPL (6d)	Full Body (3d)	Arnold (6d)	Hybrid (5-6d)	Upper/Lower (4d)
Per-muscle frequency	2x	3x	2x	2-3x	2x — sweet spot
Fatigue mgmt in deficit	Poor	Good	Poor	Moderate	Excellent
Recovery between sessions	Low	High	Low	Moderate	High
Strength retention (deficit)	Moderate	Good	Low	Moderate	Excellent
Progressive overload tracking	Moderate	Easy	Hard	Moderate	Easy
MPS coverage (≥2x/wk)					
Sustainability (you)	Hard	Volume-limited	Very hard	Hard	High

Why It Wins For You

- **Frequency that matters:** Schoenfeld et al. (2016 meta) — with volume equated, ~2x/week per muscle beats 1x; no clear benefit beyond. Upper/Lower captures this without the recovery tax of 6-day splits.
- **MPS timing:** Training stimulus elevates muscle protein synthesis ~24–48 h. Hitting each muscle every 3–4 days keeps you "topped up."
- **Strength preservation = #1 muscle-retention signal** in a deficit (Helms 2014; Roth 2022). The heavy low-rep "A" days protect your lifts.
- **Recovery is compromised on ~600 kcal deficit:** built-in rest days are the feature that lets you train hard when you do lift.
- **High existing NEAT:** 10k steps + active life means a 6-day lifting split would overreach your total systemic load.
- **Sustainability:** as an advanced lifter you need repeatable progressive overload, not novelty.

Bottom line

Optimal 2× frequency + the recovery a deficit demands + heavy work that preserves strength + sustainability for all 10 weeks. Nothing else matches that combination here.

2 Weekly Layout

MON Upper A	TUE Lower A	WED Rest + LISS	THU Upper B	FRI Lower B	SAT Opt. Day 5	SUN Rest
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Day	Session	Focus	Cardio / Steps
Mon	Upper A	Strength-leaning (heavy compounds)	10k steps PM
Tue	Lower A	Strength-leaning (squat/hinge)	10k steps PM
Wed	REST	Active recovery / mobility	10k + optional 20-30 min LISS
Thu	Upper B	Hypertrophy (volume/metabolic)	10k steps PM
Fri	Lower B	Hypertrophy (quad/glute/ham)	10k steps PM
Sat	Optional: Arms/Delts/Calves/Abs	Weak-point pump	10k + optional LISS
Sun	REST	Full rest	10k steps (easy)

Fatigue rule

The Saturday session is **optional**. Do it only when sleep, hunger, and recovery are well-managed. Drop it the moment fatigue climbs (especially Weeks 6–10). Capping sessions at 7–8 exercises is intentional to prevent overtraining in a deficit.

Timeline reconciliation

Diet/deficit runs the full **10 weeks**. Training is an **8-week mesocycle** (deload at Week 5), then Weeks 9–10 are a short peak/finish block.

3 The Workouts

Legend: RPE on Reps-in-Reserve scale (RPE 8 = 2 reps left). Tempo = eccentric-pause-concentric. Every session capped at 7–8 exercises to manage fatigue.

Day 1 — Upper A (Strength-leaning)

#	Exercise	Sets×Reps	RPE	Rest	Tempo	Cues / Progression
1	Barbell Bench Press	4×5-6	7→9	3 min	3-1-1	Scapula retracted, bar to lower sternum • double progression +2.5 kg
2	Weighted Pull-up / Pendlay Row	4×6-8	7→9	2.5 min	2-0-1	Lead with elbows, full stretch • add reps then load
3	Seated DB Shoulder Press	3×6-8	8	2 min	2-0-1	Stop at ear level • add reps then load
4	Chest-Supported Lat Pulldown	3×8-10	8	90 s	2-1-2	Drive elbows to hips • double progression
5	Incline DB Press (30°)	3×8-10	8	90 s	2-0-1	Deep stretch • double progression
6	Cable Lateral Raise	3×12-15	9	60 s	1-0-2	Lead with elbow, no shrug • add reps then load
7	EZ-Bar Curl	3×8-10	9	75 s	2-0-1	No swing, elbows pinned • double progression
8	Triceps Rope Pushdown	3×10-12	9	60 s	2-0-1	Elbows fixed, spread at bottom • double progression

Day 2 — Lower A (Strength-leaning)

#	Exercise	Sets×Reps	RPE	Rest	Tempo	Cues / Progression
1	Back Squat	4×5-6	7→9	3 min	3-1-1	Brace 360°, hit depth, drive midfoot • +2.5-5 kg
2	Romanian Deadlift	3×6-8	8	2.5 min	3-1-1	Push hips back, bar close • add reps then load
3	Leg Press	3×8-10	8	2 min	2-0-1	Control to ~90°, no slam • double progression
4	Seated Leg Curl	3×10-12	9	75 s	2-1-2	Slow eccentric, full squeeze • double progression
5	Standing Calf Raise	4×8-12	9	60 s	2-2-1	Full stretch +1s, big pause top • double progression
6	Hanging Leg Raise	3×10-15	9	60 s	2-0-1	Posterior pelvic tilt, no swing • add reps
7	Cable Crunch	3×12-15	9	45 s	2-1-2	Round spine, hips fixed • double progression

Day 3 — Upper B (Hypertrophy) — trimmed to 8 exercises

#	Exercise	Sets×Reps	RPE	Rest	Tempo	Cues / Progression
1	Incline Barbell/DB Press	4×8-10	8	2 min	2-1-1	Controlled stretch, upper-chest squeeze
2	Chest-Supported Row	4×10-12	8	90 s	2-1-1	Pause at contraction, full stretch
3	Machine Shoulder Press	3×10-12	9	90 s	2-0-1	Smooth path, no lockout rest
4	Wide-Grip Lat Pulldown	3×10-12	9	75 s	2-1-2	"Elbows to floor", lat stretch up top
5	Cable Fly / Pec Deck	3×12-15	9	60 s	2-1-1	Deep stretch, hug + squeeze
6	Reverse Pec Deck / Face Pull	3×15-20	9	45 s	1-1-1	Pull to forehead, rotate pinkies up (rear delt)
7	DB Lateral Raise	3×12-20	10	45 s	1-0-2	Slow lower, constant tension
8	Incline DB Curl superset Overhead Triceps Ext	3×10-12 ea	9	60 s	2-1-1	Time-efficient arm finisher

Day 4 — Lower B (Hypertrophy)

#	Exercise	Sets×Reps	RPE	Rest	Tempo	Cues / Progression
1	Hack Squat / Pendulum	4×10-12	8	2 min	3-1-1	Deep ROM, quads loaded
2	Barbell/Machine Hip Thrust	3×10-12	8	90 s	2-1-1	Chin tucked, full glute lockout +1s
3	Bulgarian Split Squat	3×10-12/leg	9	75 s	2-0-1	Slight torso lean, front-leg drive
4	Lying Leg Curl	3×12-15	9	60 s	2-1-2	No hip rise, slow eccentric
5	Leg Extension	3×12-15	10	60 s	2-1-1	1s squeeze at top
6	Seated Calf Raise	4×12-15	9	45 s	2-2-1	Soleus focus, full stretch
7	Standing Calf Raise	3×12-15	9	45 s	1-1-1	Bouncy stretch-reflex reps
8	Weighted Decline Crunch / Ab Wheel	3×12-15	9	45 s	2-1-1	Flex spine, exhale at top

Day 5 (Optional) — Arms / Delts / Calves / Abs

#	Exercise	Sets×Reps	RPE	Rest
1	DB/Cable Lateral Raise	4×15-20	10	45 s
2	Reverse Pec Deck (rear delt)	3×15-20	10	45 s
3	Cable Curl (superset #4)	3×12-15	10	30 s
4	Cable Triceps Pushdown	3×12-15	10	30 s
5	Standing Calf Raise	4×12-20	10	45 s
6	Hanging Leg Raise + Cable Crunch	3×15 ea	10	45 s

4 Weekly Volume vs. Evidence

Hard sets/week (direct + meaningful indirect), Day 5 included. Anchor: Schoenfeld 2017 dose-response — 10+ sets/muscle/week drives growth; in a deficit, sit near the lower-effective end (MEV–MAV) to preserve muscle without exceeding recovery.

Muscle	Your Sets/wk	Evidence Target	Verdict
Chest	13-15	10-18	Optimal
Back (overall)	14-16	12-20	Optimal
Lats	8-10	8-14	Good
Upper back	6-8	6-12	Good
Front delts	9-11	6-12	Covered by pressing
Lateral delts	10-13	12-22	Solid
Rear delts	6-8	6-12	Good
Biceps	9-12	8-16	Optimal
Triceps	10-13	10-16	Optimal
Forearms	grip + 2-4	4-10	Adequate
Quads	11-13	10-18	Optimal
Hamstrings	9-11	8-16	Optimal
Glutes	8-10	6-16	Good
Calves	11-14	10-20	Strong
Abs	9-11	6-16	Good

Read-out

Every muscle sits at or above the 10-set growth threshold and within recoverable range for a dieting advanced lifter. Trimming Upper B from 11 to 8 exercises removed only redundant arm/forearm sets — weekly volume stays optimal while you train fresher.

5 Progressive Overload (8 + 2 Weeks)

Core method = **Double Progression with RPE auto-regulation**. Add reps within a range until you top it on all sets, then add load (2.5 kg upper / 2.5–5 kg lower) and drop to the bottom.

Week	Phase	Intensity (RPE)	Volume	Action
1	Accumulation 1	7 (3 RIR)	Base	Establish loads, perfect technique
2	Accumulation 1	8 (2 RIR)	Base	Add reps across sets
3	Accumulation 1	8-9	+1 set lagging	Add reps; load up where maxed
4	Accumulation peak	9 (1 RIR)	Peak	Push compounds to top of range
5	DELOAD	5-6 (-40-50% load)	Half	Recover, movement quality, refeed
6	Intensification	8	Base	Reset heavier than W3, rebuild reps
7	Intensification	8-9	+1 set lagging	Add reps/load
8	Intensification peak	9 (1 RIR)	Peak	Top-set PRs where possible
9	Peak / Maintain	8-9	Slightly reduced	Hold strength, maximize definition
10	Peak / Finish	7-8 (taper)	Reduced	Maintain intensity, drop junk volume

In a deficit, success = **MAINTAINING** loads, not weekly PRs

Holding your numbers while losing 5–7 kg means you're getting relatively stronger and preserving muscle. That IS progress.

6 Nutrition

Maintenance Calories — Method Comparison

Method	BMR	×1.55	BF	LBM	Katch BMR
Mifflin-St Jeor	1,763	~2,730	20%	64.8 kg	1,770
Katch-McArdle (22% BF)	1,735	~2,690	22%	63.2 kg	1,735
			25%	60.8 kg	1,682

Maintenance ≈ 2,700 kcal/day

Activity ×1.55 (4-5 lifts + 10k steps + active life). Both methods agree → high confidence. Verify against 2-week weight trend.

Fat-Loss Calories

Approach	Deficit	Intake	Loss/wk	Muscle risk
Moderate	-15% (-400)	2,300	~0.4 kg	Very low (too slow)
Moderate-Aggressive	-22% (-600)	2,100	~0.6 kg	Low
Aggressive	-30% (-800)	1,900	~0.8 kg	Moderate-high

Chosen: ~2,100 kcal/day

Helms 2014 / Garthe 2011: ~0.5–0.7%/week BW loss maximizes fat loss while sparing muscle. ~0.6 kg/wk × 10 wks = ~6 kg → ~75 kg, landing in target while moderate enough (with high protein + heavy lifting) to protect muscle.

Macronutrients @ 2,100 kcal

Macro	Amount	g/kg	kcal	%	Rationale
Protein	190 g	2.35	760	36%	Helms: 2.3-3.1 g/kg LBM in deficit; preserves muscle + satiety
Fat	60 g	0.74	540	26%	Hormones, joints, EFAs
Carbs	200 g	2.5	800	38%	Training performance + recovery
Fiber	30-40 g	—	—	—	Satiety, gut health

Hydration & Electrolytes

Water	Sodium	Potassium	Magnesium
3-3.5 L/day	3,000-4,500 mg	3,500-4,700 mg	350-400 mg

Protein Distribution & Timing

- **5 meals**, every 3-4 h — ~0.4 g/kg/meal (Schoenfeld & Aragon 2018) maximizes daily MPS.
- **35-45 g/meal** (≥3 g leucine to cross the leucine threshold).
- **Pre-WO (AM):** ~25 g whey ± banana. **Post-WO:** ~40 g protein + carbs within 1-2 h.
- **Before bed:** ~30-40 g slow protein (paneer/Greek yogurt/milk) — Res 2012, overnight MPS.

Sample Day — ~2,100 kcal | ~188 P / 58 F / 200 C

Meal / Time	Foods & Portions	P	F	C	kcal
Pre-WO 6:00	1 scoop whey + black coffee	24	2	2	120
Breakfast 8:30	50 g oats + 1 scoop whey + 150 g berries + 150 ml skim milk	40	7	55	430
Lunch 1:00	150 g tofu (or 120 g low-fat paneer) + 1 cup dal + 120 g rice + veg (1 tsp oil) + salad	33	16	65	540
Snack 4:30	200 g low-fat Greek yogurt + 15 g almonds	24	12	12	250
Dinner 8:00	55-60 g dry soya chunks (cooked) + 2 wheat rotis + large veg (1 tsp oil) + salad	39	11	56	480
Bed 10:00	100 g low-fat paneer or 200 g Greek yogurt	18	8	6	160
TOTAL	If short on protein, add ½ scoop whey (+12 P)	~188	~56	~196	~2,000-2,100

Meal Alternatives

Role	Options
Lean protein	Paneer, tofu, soya, tempeh, Greek yogurt, whey, skim milk
Legumes	Dal, rajma, chickpeas, beans, moong
Carbs	Oats, rice, roti, potato, sweet potato, fruit, quinoa
Fats	Almonds, walnuts, chia/flax, peanut butter, ghee/olive oil
Veg	Spinach, broccoli, capsicum, bhindi, cauliflower

Budget Swaps

Premium	Budget
Tofu/tempeh	Soya chunks (cheapest)
Greek yogurt	Hung curd
Whey isolate	Whey concentrate / milk+paneer
Almonds	Peanuts / roasted chana
Berries	Banana, papaya, seasonal fruit
Low-fat paneer	Homemade from toned milk

Protein-quality tip

Combine legumes + grains (dal+rice, rajma+roti) across the day for complete amino acids. Soy, dairy & whey are already complete.

8 Supplements

Supplement	Status	Dose	Timing	Why
Creatine Monohydrate	ESSENTIAL	5 g/day	Any time, daily	Strongest evidence; vegetarians respond more
Whey Protein	ESSENTIAL	1-2 scoops	Pre/post WO	Convenient complete protein to hit 190 g
Vitamin B12	ESSENTIAL	250-500 mcg/day	With food	Not reliable in veg diet; nerves/blood/energy
Vitamin D3 (lichen)	ESSENTIAL	1,000-2,000 IU	With fatty meal	Common deficiency; bone/immune/performance
Omega-3 (Algae EPA/DHA)	RECOMMENDED	0.5-1 g EPA+DHA	With food	Veg diets low in EPA/DHA; recovery/joints/heart
Magnesium	OPTIONAL	200-400 mg	Evening	Often low in deficit; sleep/muscle
Electrolytes	OPTIONAL	As needed	Around training	High steps + sweat + low-sodium veg diet
Caffeine	OPTIONAL	3-6 mg/kg	30-45 min pre-WO	Performance, focus, appetite blunting
Citrulline Malate	OPTIONAL	6-8 g	Pre-WO	Modest pump/endurance
Beta-Alanine	OPTIONAL	3-5 g/day	Any time, daily	Helps 60-240 s high-rep efforts

Verdict

Essentials = Creatine, B12, D3, whey (practical), algae omega-3. Everything else is optional polish. No fat burners — they don't move the needle.

9 Cardio

Modality	Fat loss	Muscle/recovery cost	Verdict
Walking (NEAT)	High	Negligible	Keep 10k steps — your best tool
Incline treadmill (LISS)	High	Low	Primary structured cardio
Cycling (LISS)	Good	Low	Good alternative
HIIT	High/min	High	Max 1×/wk, away from leg days

- **Keep your 10,000 evening steps** — non-negotiable, your single best fat-loss tool. Don't raise it first; save step increases for plateaus.
- **Add 2-3 LISS** sessions, 20-30 min, after lifting or on rest days.
- **Fasted cardio not superior** (Schoenfeld 2014, matched calories). Walking fasted is fine; for lifting take ~25 g whey first.

10 Recovery

Sleep (biggest recovery lever)

- Target 7.5-9 h; push toward 8+.
- Fixed times, dark/cool room (~18-20°C), no screens 60 min prior; cap caffeine early.

Warm-Up (RAMP, 8-10 min)

- 3-5 min easy cardio
- Dynamic mobility for the day's joints
- 2-3 ramp-up sets on first compound

Cool-Down

- 5 min easy walk + nasal breathing + light static stretch

Mobility (10 min, 3×/wk)

- Hip flexor, 90/90, deep squat hold, T-spine rotations, shoulder dislocates, hamstring/calf stretch

Foam Rolling

- Pre-training, tight areas 30-60 s each (warm-up adjunct)

Stress

- 10 min/day decompression — chronic stress + deficit = high cortisol, water retention masking fat loss

Recovering Well

Stable/rising lifts; good sleep; normal RHR; steady mood; normal appetite

Back Off / Deload

Strength dropping 2 sessions; RHR +5-10 bpm; poor sleep; persistent fatigue; joint aches; soreness >3 days

11 Tracking

Metric	How	Frequency
Body weight	Same time AM, fasted; record 7-day average	Daily → weekly avg
Waist	Tape at navel, relaxed, AM fasted	Weekly
Photos	Front/side/back, same lighting/poses	Weekly
Strength	Log load × reps × RPE	Every session
Body fat %	Calipers or visual vs. photos	Every 2-3 weeks

Calorie adjustment rule

Compare weekly averages. Losing 0.4-0.7 kg/wk → change nothing. >0.8 kg/wk for 2 wks → +150 kcal. <0.3 kg/wk for 2 wks → go to plateau tree. Waist + photos override the scale.

12 Plateau Strategy

Confirm a real stall first: weekly **average** weight flat for 2 weeks AND waist + photos unchanged. Water retention often masks fat loss for 1-2 weeks — don't panic early.

START Real stall confirmed (2 wks: weight + waist + photos all flat)

1 AUDIT FIRST — re-log food 3 days honestly (portion creep!). Hitting steps + sleep + protein? Fix adherence before changing the plan. *Most "plateaus" are this.*

2 ↑ **NEAT** — 10k → 12k steps or +1 LISS. Lowest cost to recovery; preserves muscle.

3 ↓ **Calories** modestly — cut ~150 kcal from carbs/fat (never protein). 2,100 → ~1,950.

4 + **Cardio** — add 1 LISS session (20-30 min). Keep HIIT ≤1×/wk.

5 **Check training** — if fatigued, REDUCE junk volume, don't add. Fatigue ≠ fat loss.

6 **Refeed / Diet break** — 1-2 days (or 7-14 days) at maintenance to restore leptin/performance & restart fat loss (Byrne 2018 MATADOR).

Re-evaluate after 2 weeks → repeat from top if needed.

Golden rules

Protein and lifting intensity are the LAST things you cut. Move NEAT/cardio and small calorie nudges first. A planned refeed/diet break is a tool, not a failure.

13 Expected Results (10 Weeks)

Metric	Start	Realistic Outcome	Notes
Body weight	81 kg	74-76 kg (-5 to -7)	~0.6 kg/wk
Body fat %	~22%	~14-16%	Visible abs, separation
Fat mass	~18 kg	~11-12 kg (-6 to -7)	Bulk of the loss
Lean body mass	~63 kg	maintain ± small gain	Preserve = the win
Waist	baseline	-5 to -8 cm	Strongest visual change
Strength	baseline	Maintained	Holding loads = success
Appearance	soft ~22%	Lean, athletic, vascular, defined	Dramatic visual transformation

The honest truth

As an advanced lifter in a deficit, large simultaneous muscle gain is unlikely (that's a beginner phenomenon). Your excellent, achievable outcome: lose ~6-7 kg fat, preserve essentially all muscle, possibly add a little. Far leaner → you'll LOOK much more muscular as hidden muscle becomes visible. 100% achievable naturally.

Daily

- ~2,100 kcal / ~190 g protein
- 5 protein-spread meals (incl. pre-bed slow protein)
- 3-3.5 L water + salt food
- 10,000 steps
- Creatine 5 g + B12 + D3 + algae omega-3
- 7.5-9 h sleep
- Log workout (load × reps × RPE)

Weekly

- 4 lifts (+ optional Day 5)
- 2-3 LISS sessions
- Weigh-in: 7-day average
- Waist + progress photos
- Review: progressing? losing ~0.6 kg? adjust
- Mobility 3× + recovery check

Phase Milestones

- **Week 5:** Deload (light, recover, refeed)
- **Weeks 6-8:** Intensification — chase rep/load PRs where recovery allows
- **Weeks 9-10:** Peak/finish — maintain intensity, trim junk volume, lock in the lean look

Your complete protocol

Single, fully-optimized plan for your 174 cm / 81 kg advanced vegetarian profile, morning training, high step count, and 10-week lean-out goal. Execute the levers in order, let weekly averages guide adjustments, keep the heavy work heavy, and protect sleep + protein above all. 15% body fat at ~75 kg is well within reach.

This document provides general evidence-based fitness and nutrition information, not personalized medical advice. As a long-term vegetarian entering a sustained deficit, consider a baseline blood panel (vitamin D, B12, ferritin, lipids) and consult a qualified professional before making significant changes.